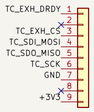
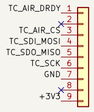
SIROKKO 262.01 — CONTROLLER
YD-ESP32-S3 N16R8 V1.4: 24 V supply.
Motor: external HC-160A-S2: K1 glow; K2 fuel clutch.

THERMOCOUPLES: J30 EXHAUST / J31 HOT AIR — TYPE K
9-pin MAX31856 modules, Adafruit 3263 pin order and geometry.
VIN=3V3; 3Vo/FLT not connected; probes connect at module terminals.

J30
MAX31856 EXHAUST




J31
MAX31856 AIR

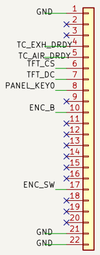



J1




DEVKITC1_J1

J3




DEVKITC1_J3

LM5013 supplies +5V_DEVKIT.
The installed DevKit is powered only from Sirokko.
Remove the DevKit before connecting USB for programming.

Power5V




File: Power5V.kicad_sch

MotorInterface

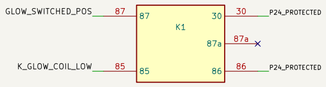



File: MotorInterface.kicad_sch

J32




TFT_2.4_EC11_SPI_3V3_100mm_MAX




DG858M-5021-76-1024 + DZ85AB-5-PCB / GLOW




DG858M-5021-76-1024 + DZ85AB-5-PCB / CLUTCH_RUN

J12




ROOM_DS18B20_HARNESS

TWO RELAYS: K1 GLOW; K2 CLUTCH. U11 IN1=GLOW_CMD, IN2=FUEL_CMD.
IN3/IN4, nENBL and RESET tied to GND.
No independent series contact for a welded K2; motor control is external.

MAIN INPUT / ROOM SENSOR
J13: 1=VIN_RAW_24, 2=GND. Design supply 21.6…28.8 V.
DevKit is powered only from this board; remove it before USB programming.
J12: 1=3V3, 2=DQ, 3=GND. Never apply 24 V to logic terminals.

J13



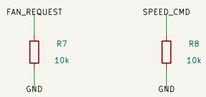

24V_INPUT_PHOENIX_1756993

InputProtection




File: InputProtection.kicad_sch



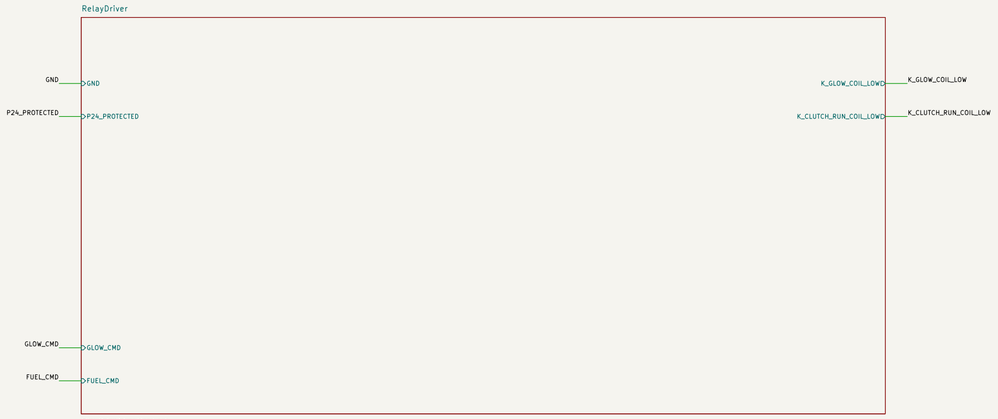

PANEL: SPI, 3.3 V, TFT + EC11 + KEY0; J32 IDC 2x6, NOT A 1:1 CABLE.
J32 -> module: 1->1 GND; 2->2 VCC; 3->4 MOSI GPIO7; 5->3 SCK GPIO8;
6->9 A GPIO5; 7->10 B GPIO38; 8->11 PUSH GPIO47;
9->5 RES GPIO6; 10->6 DC GPIO41; 11->7 CS GPIO42; 12->12 KEY0 GPIO40.
J32.4=GND; module pin8 BLK not connected; panel must support default-on backlight.
PANEL_SDA/PANEL_SCL are SPI, not I2C. Cable <=100 mm.
KEY0 must close to GND; long press starts/stops. Encoder handles the menu.

CONTROLLER POWER AND RESET
J1.6/GPIO6 drives TFT_RES on J32.9.
No external watchdog/latch or independent 24 V presence input.
Firmware disables fuel/glow after reset and requires an explicit start.

RELAY DRIVER — DRV8803
IN1=GLOW_CMD; IN2=FUEL_CMD. Unused IN3/IN4, nENBL and RESET=GND.
OUT1 sinks K1 coil; OUT2 sinks K2 coil. OUT3/OUT4/nFAULT not connected.

DEFINED INPUTS / OUTPUTS
R1: room-sensor pull-up; R2/R3: thermocouple CS pull-ups.
R7/R8: fan/PWM pulldowns; relay pulldowns on RelayDriver sheet.
The external TFT 2.4 + EC11 + KEY0 panel connects to J32.

RelayDriver




File: RelayDriver.kicad_sch

J28




CLUTCH_OUTPUT_PHOENIX_1756993

J27 GLOW: 1=switched positive; 2=power GND.
Design load: 7 A continuous; 10 A for up to 1 s.

J28 CLUTCH: 1=switched positive; 2=power GND.
Design load: 2 A continuous; 3 A for up to 1 s.

LOCAL PANEL
TFT 2.4 + EC11 + KEY0 at J32; KEY0=START/STOP, GPIO40.
No separate user controls on the carrier.
The DevKit retains its own EN/BOOT buttons and USB programming interface.

J27



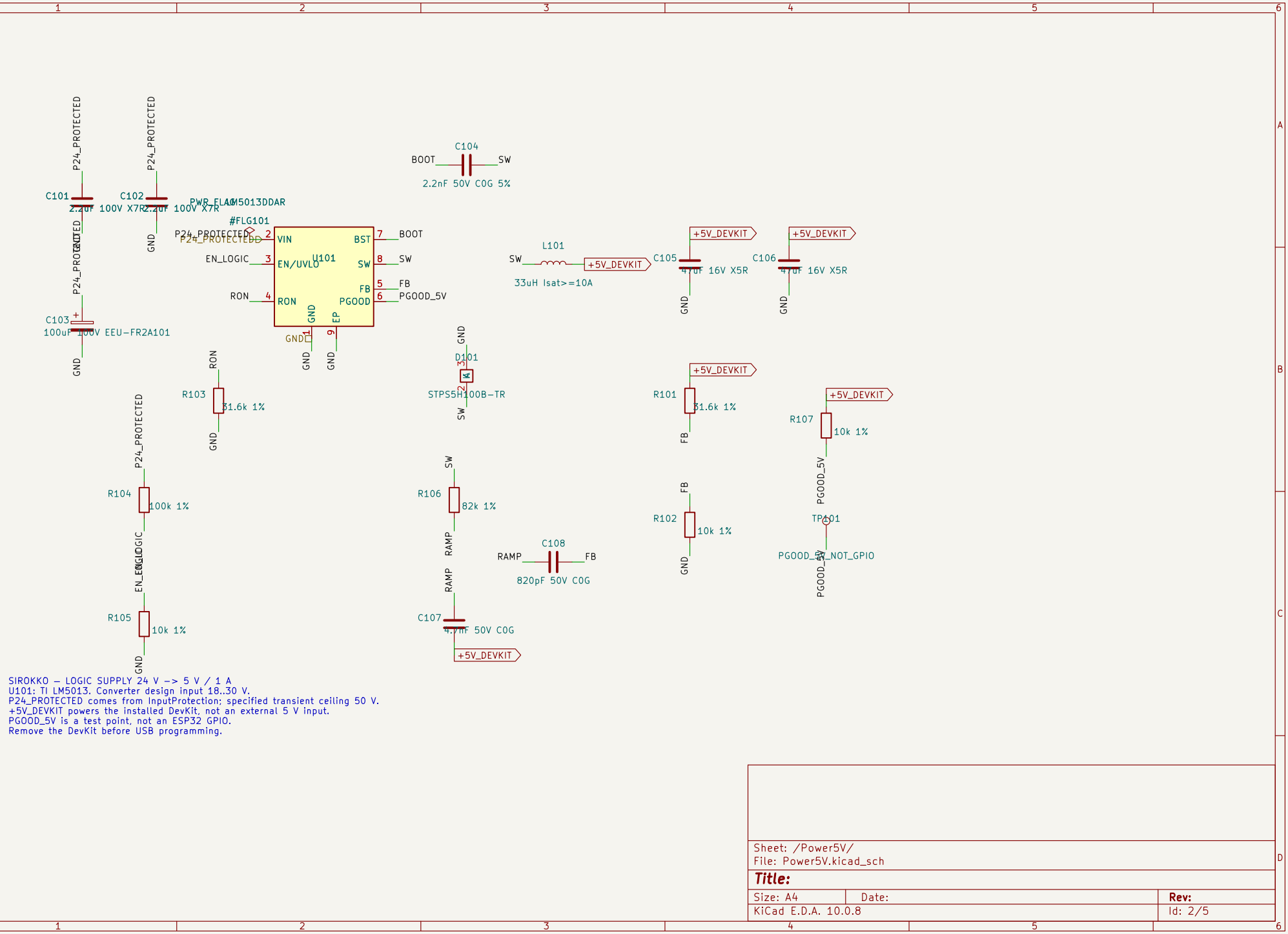

GLOW_OUTPUT_PHOENIX_1756993

24 V heater controller

Sheet: /
File: Sirokko.kicad_sch

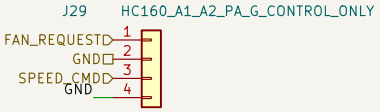
**Title: Sirokko 262.01 — common controller interfaces**

| Size: A2 | Date: 2026-10-23 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 1/5 |




SIROKKO — LOGIC SUPPLY 24 V -> 5 V / 1 A
U101: TI LM5013. Converter design input 18..30 V.
P24_PROTECTED comes from InputProtection; specified transient ceiling 50 V.
+5V_DEVKIT powers the installed DevKit, not an external 5 V input.
PGOOD_5V is a test point, not an ESP32 GPIO.
Remove the DevKit before USB programming.

| | |
|---|---|
| Sheet: /Power5V/ | |
| File: Power5V.kicad_sch | |
| **Title:** | |
| Size: A4 | Date: |
| KiCad E.D.A. 10.0.8 | Rev: Id: 2/5 |



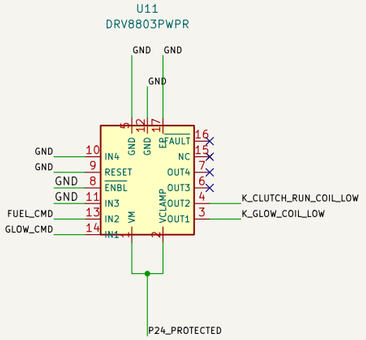

| | | |
|---|---|---|
| Sheet: /MotorInterface/ | | |
| File: MotorInterface.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.8 | | Id: 3/5 |

SIROKKO — RELAY DRIVER

24 V RELAY OUTPUTS — DRV8803

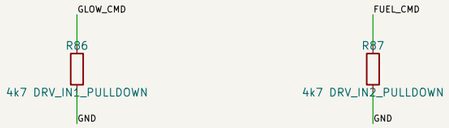
Commands are active HIGH; relay coils are switched on the low side.

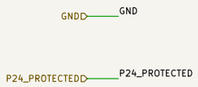
IN1=GLOW_CMD; IN2=FUEL_CMD. R86/R87 keep commands LOW.
Unused IN3/IN4 and nENBL/RESET=GND; OUT3/OUT4 not connected.
No independent welded-contact cutoff.
Firmware: internal ESP32-S3 IWDT/TWDT, supervised control task,
outputs OFF after reset, no automatic restart.

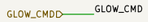



ESP32 controls startup, purge, operating timing and faults. Motor power is external.










K1: glow plug. K2: fuel-pump clutch. Both use normally-open contacts.

| | | |
|---|---|---|
| Sheet: /RelayDriver/<br>File: RelayDriver.kicad_sch | | |
| **Title:** | | |
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.8 | | Id: 4/5 |

INPUT PROTECTION
F1: Littelfuse 0997015.WXN, 15 A / 58 V DC, Keystone 3568 holder.
LM74800 + MOSFETs: reverse polarity / power path; TVS3301: transient clamp.

VIN_RAW_24
P24_PROTECTED
F1
0997015.WXN / 15A
C24
220nF 100V X7R 10% CL21B224KCFSFNE
D1
TVS3301DRBR WIP
GND
U13
LM74800QDRRRQ1 WIP_NOT_TRANSIENT_QUALIFIED
#FLG0101
PWR_FLAG
R90
4.7R Q1_GATE
R91
4.7R Q2_GATE
R92
4.7R Q3_GATE
R93
4.7R Q4_GATE
Q1
Q2
Q3
Q4
IPB017N10N5LFATMA1 WIP
R94
84.5k 1% OV_TOP
R95
3.48k 1% OV_BOT
R96
75k 1% EN_TOP
R97
5.62k 1% EN_BOT
C25
1uF 100V CVS GRM31CR72A105KA01L
REV_MID: intermediate MOSFET power-path rail; PWR_FLAG declares the U13.VS source.

INPUT SLEW CONTROL: R108 + C109, TI LM7480 Fig. 9-3.
C109=33 nF. This circuit controls the voltage rise, not the load current.
R108
22k 1% CdVdT isolation WIP
C109
33nF 100V C0G 5% CdVdT WIP

Sheet: /InputProtection/
File: InputProtection.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.8
Id: 5/5